

The Sun in A Chip

Era of Solar Devices Not Too Far

With advancements in photovoltaic, material and chemical sciences, and notable leaps in semiconductor manufacturing technologies, solar cells are definitely gaining ground. Yet, they are unable to compete with other energy sources for fuelling mainstream devices. But hope is in sight



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The process of solar power generation is simple. The photons in sunlight are absorbed by thin wafers or sheets of a semiconductor material such as crystalline silicon; electrons are released; these flow through the material and produce a small amount of electricity. However, an array or panel of solar cells can produce a reasonable amount of direct current. Semiconductor manufacturers can easily produce solar cells following

the procedures used to manufacture computer chips, with different equipment. So, what is stopping the world from moving to solar power?

Several industry majors like National Semiconductors, Sharp, Moser Baer, Applied Materials India, Tata BP and GE are involved in this space. A range of solar-powered equipment is available in the market. And, solar power is beginning to gain ground.

"There is a huge demand for locally manufactured products, which presents bigger opportunities for domestic solar solutions. There are

diverse and innovative solar products available in the market, such as solar lanterns, solar home lights, solar-powered water pumps, grain grinders, crop dryers and Internet communication systems. These products are energy-efficient and cost-effective," says Dr Madhusudan Atre, president, Applied Materials India.

We are all familiar with solar-powered calculators and watches. Many a rooftop is decked with a solar panel, either for heating water or for running a few appliances. Often we read about solar-charged laptops, mobile phones, vehicles, and even 'green' buildings powered entirely by solar energy. Besides, solar cell arrays contribute electricity to power grids. Yet, solar power still has not been accepted as the mainstream energy source. The reasons are simple—solar cells are still unable to match the cost and efficiency of other fuels and power sources. Besides, the lack of awareness about solar energy and solar-powered products is hampering the technology from attaining critical mass.

Presently, there is a lot of research happening to find the ideal semiconductor, maximise the power conversion efficiency of the cell, figure out how the current generated can be put to more uses, improve the life of the solar cells, and so on. As these ventures pay off and as awareness spreads, solar energy will definitely gain ground because it is the most environment-friendly and abundant fuel one can possibly think of!